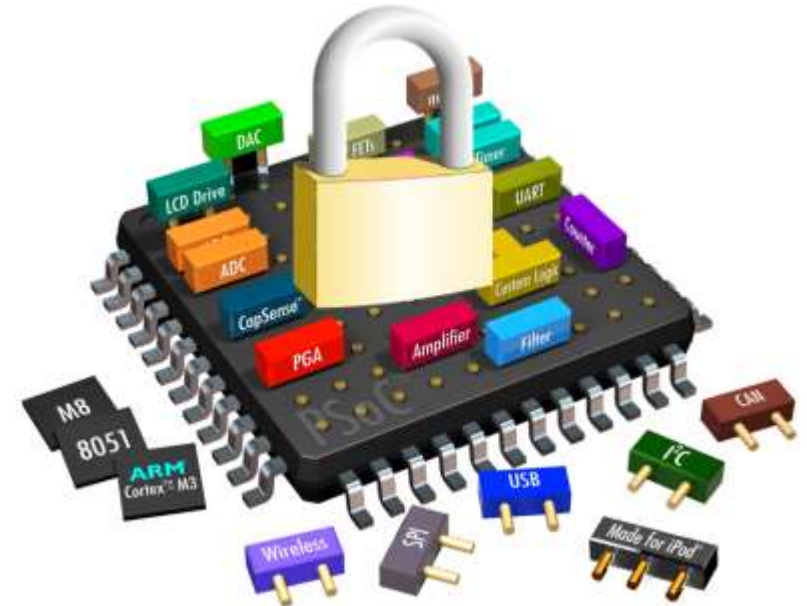


Hardware Security

TEE Examples: AMD SEV



SEV Evolution

1**2016: SEV (Secure Encrypted Virtualization)**

First x86 technology to isolate VMs from hypervisor using main memory encryption. Each VM gets unique AES encryption key for automatic data protection.

2**2017: SEV-ES (Encrypted State)**

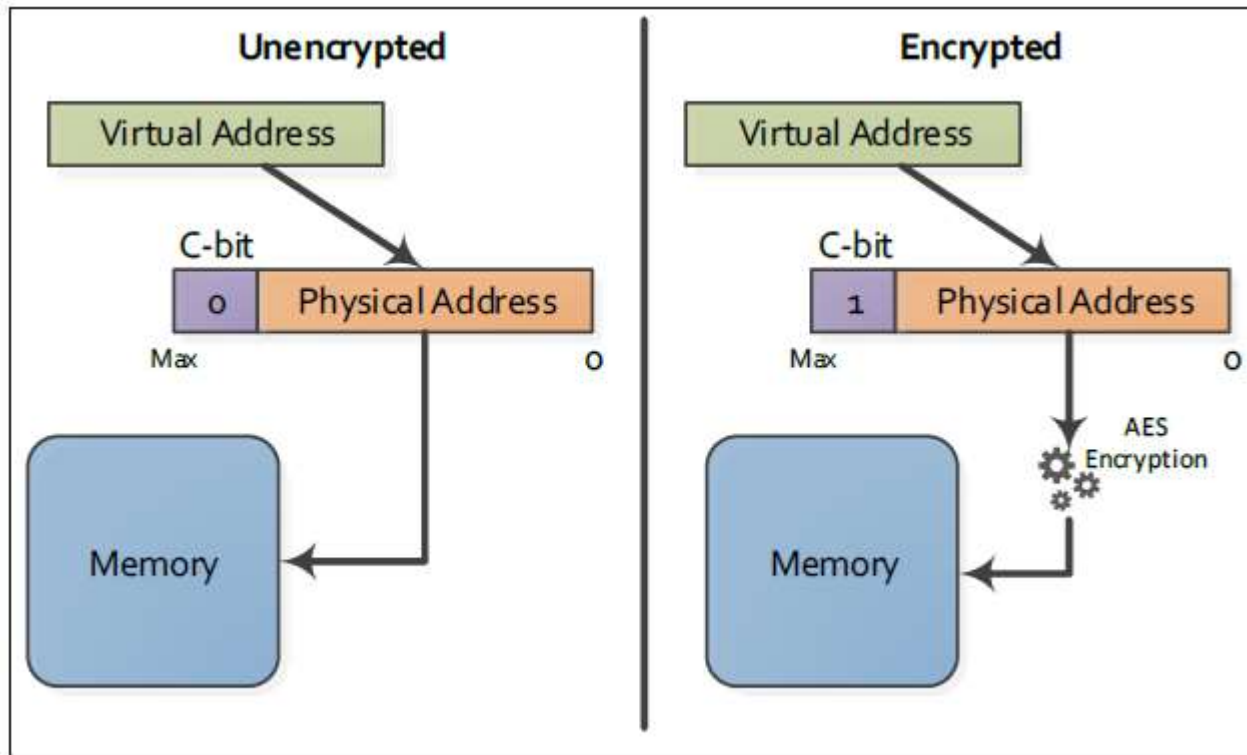
Added protection for CPU register state. VM register state encrypted on each hypervisor transition, preventing hypervisor from seeing actively used data.

3**2020: SEV-SNP (Secure Nested Paging)**

Introduces strong memory integrity protection to prevent malicious hypervisor attacks like data replay, memory re-mapping, and corruption.



Private vs. Shared Memory



- VMs control memory page status using the enCRypted bit (C-bit) in guest page tables. The C-bit location is implementation-defined and may be the top physical address bit.
- This architecture enables VMs to selectively share data while maintaining strong protection for private information.

Threat Model

■ Trusted Components

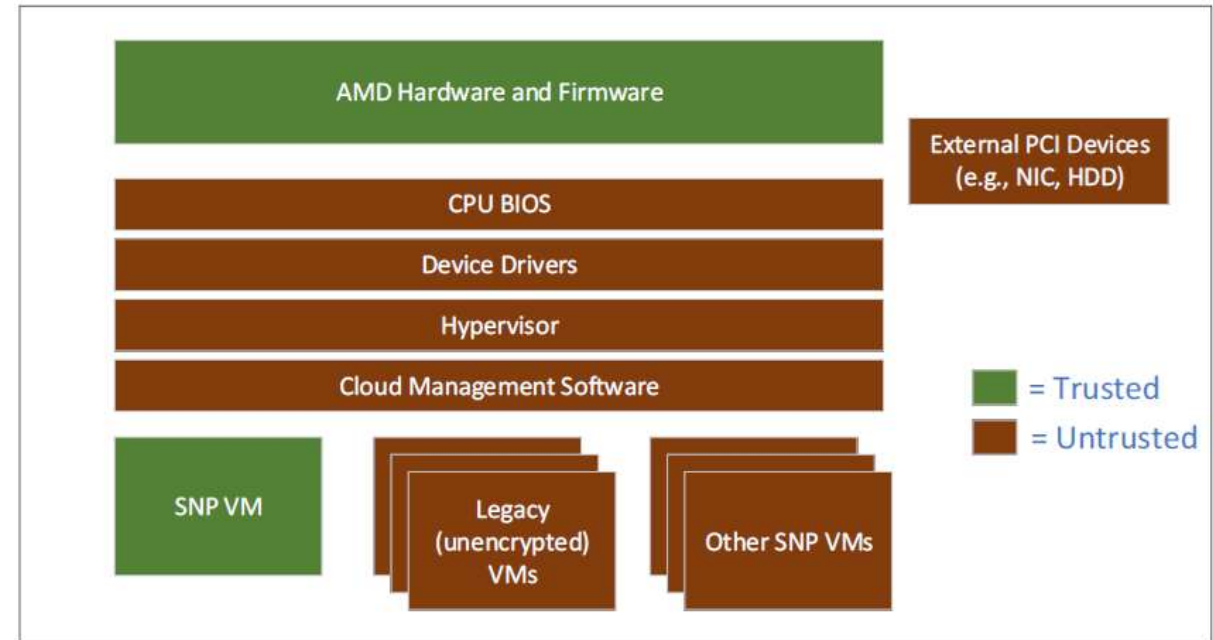
- AMD System-On-Chip hardware, AMD Secure Processor, and the VM itself are fully trusted. VM must protect its own interfaces following security best practices.

■ Untrusted Components

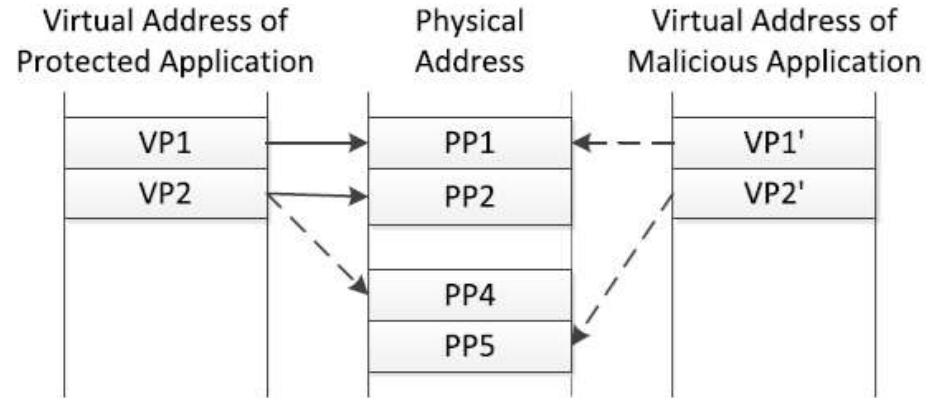
- All other CPU software and PCI devices treated as fully untrusted including BIOS, hypervisor, device drivers, and other VMs, assumed potentially malicious.

■ Protection Scope

- SEV-SNP protects against confidentiality and integrity attacks, maintains hypervisor availability, and includes TCB rollback protection and optional interrupt/side-channel defenses.



Memory Mapping Threats



- Data Corruption
 - Only the owner of a memory page can write that page
- Memory Aliasing
 - Every physical memory page can map only to a single guest page at one time
- Memory Re-mapping
 - Every guest page can map only to a single physical memory page at one time

Reverse Map Table

■ Inverted Page Table

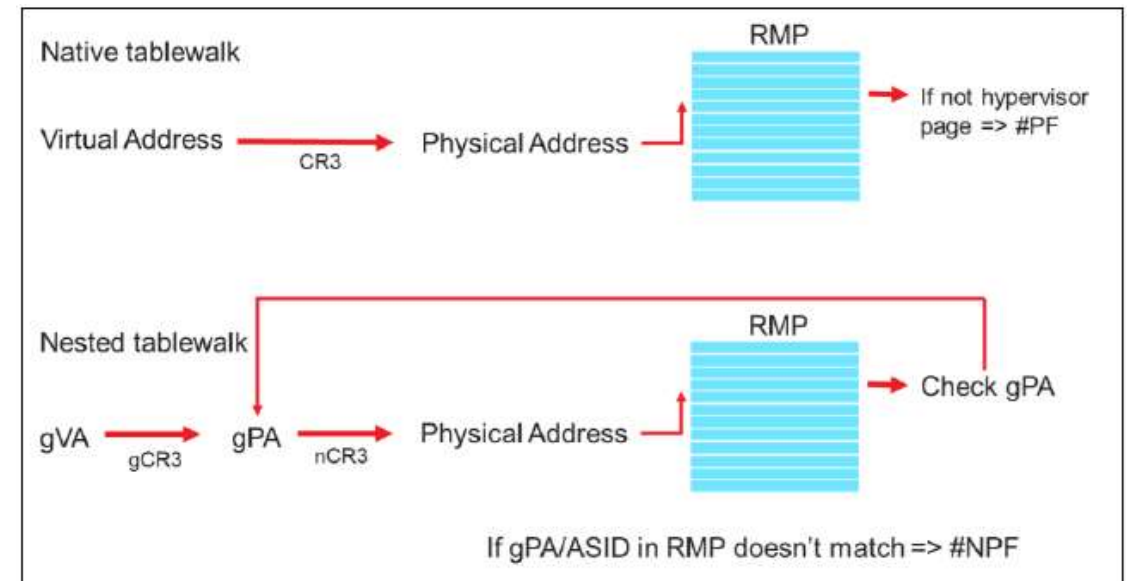
- Mapping physical to virtual addresses

■ Native Mode

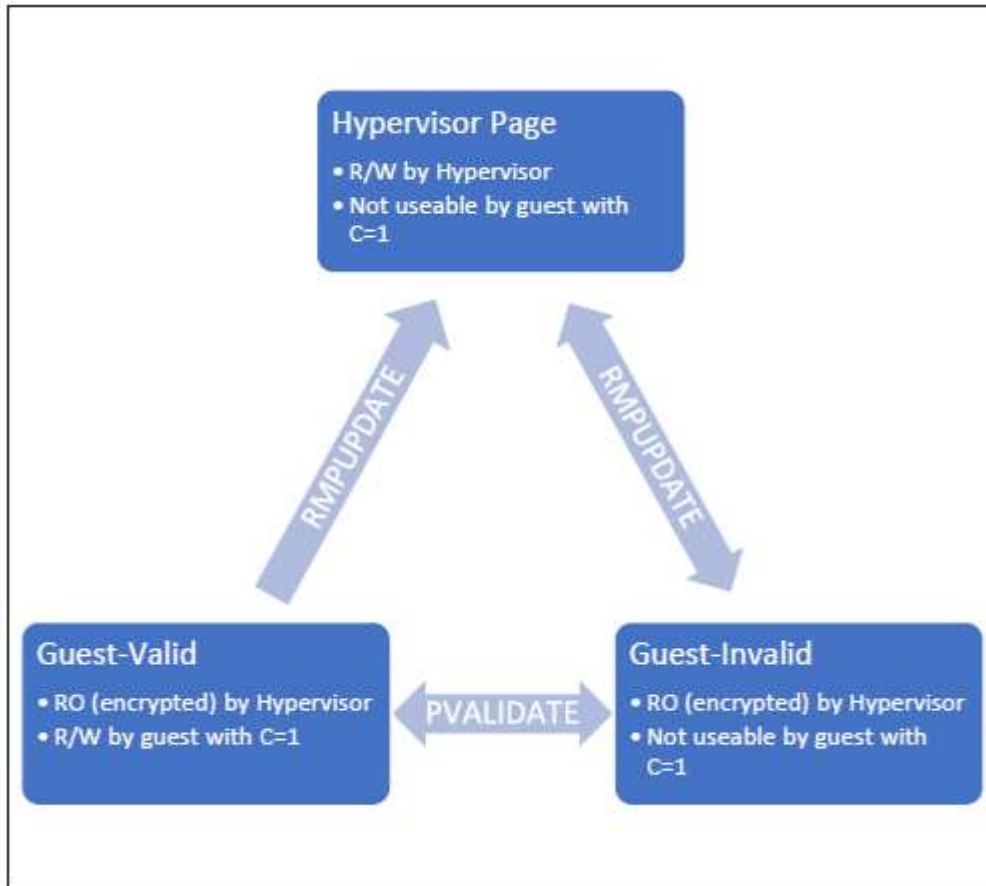
- Virtual addresses translated to physical addresses using standard x86 page tables. Final physical address indexes RMP. If page is hypervisor-owned, checks pass and TLB entry created. Otherwise, table-walk faults.

■ SEV-SNP VM Mode

- Guest Virtual Address translated to Guest Physical Address, then to System Physical Address using 2-level paging. SPA indexes RMP. Entry must show page is guest-owned, assigned to this specific guest, and mapped at correct GPA.



Page Validation



Each RMP entry contains a Validated bit, automatically cleared to 0 when new RMP entry is created for a guest. Pages with Validated bit clear cannot be used as private guest pages until the guest sets the bit using the PVALIDATE instruction.

Hypervisor Assignment

Hypervisor assigns page to guest using RMPUPDATE instruction, transitioning page to Guest-Invalid state

Guest Validation

Guest validates page using PVALIDATE instruction, transitioning to Guest-Valid state where it can be used

One-Time Rule

Guest VM should never validate same GPA more than once, ensuring injective mapping between GPAs and SPAs

Page Re-mapping Attack

